



Activity-based Project Report
on
**DATA-DRIVEN INVESTMENT RISK PREDICTION USING MACHINE
LEARNING AND SHAP-BASED INTERPRETABILITY**

Minor track project report

Submitted to Vishwakarma University, Pune

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CERTIFICATE

This is to certify that **Mr. Ayan Shaikh** (PRN:3123032010026) has completed his/her **Minor track Project** entitled “**Data-Driven Investment Risk Prediction Using Machine Learning and SHAP-Based Interpretability**” at **Vishwakarma University** satisfactorily as a partial fulfillment of the requirement for the **Bachelor’s in Computer Science (BSc-CS)** program of **Vishwakarma University, Pune** during the academic year **2025-26**.

Project Guide Sign

Internal Examiner

External Examiner

GUIDE CERTIFICATE

This is to certify that **Ayan Shaikh** a student of Vishwakarma University, Pune, has successfully completed her Minor Track Project titled “**Data-Driven Investment Risk**”

Prediction Using Machine Learning and SHAP-Based Interpretability” under my guidance during the prescribed academic period.

The project work has been carried out satisfactorily and meets the academic standards, requirements, and evaluation criteria laid down by the University for the Minor Track Project. The work submitted is original and is the result of the student’s own efforts under proper guidance.

To the best of my knowledge, the project has not been submitted previously to this or any other University or Institution for the award of any degree, diploma, or certificate.

Hence, the project work is recommended for acceptance.

Dr. Karthika P. Devan

Department: Commerce and Management

University: Vishwakarma University, Pune

Signature of the Guide

DECLARATION

I am **Ayan Shaikh**, a Bonafide student of **Vishwakarma University** , pursuing Bachelor’s in Computer Science, do hereby declare that the study entitled “**Data-Driven Investment Risk Prediction Using Machine Learning and SHAP-Based Interpretability**” is my authentic work, I have completed my study under the guidance of **Dr. Karthika P. Devan** , **Vishwakarma University**, Pune . All the data furnished in this project report are authentic and

genuine and this report neither full nor in part has ever been submitted for award of any other degree to either this university or any other university.

Student Name: Ayan Shaikh

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
SHAP	SHapley Additive exPlanations
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
DFL	Degree of Financial Leverage
EPS	Earnings Per Share

LIST OF STATISTICAL TOOLS USED

Sr. No.	Statistical Tools Used
1	Python Programming Language
2	Pandas
3	NumPy
4	Scikit-learn
5	Matplotlib
6	SHAP Library
7	Google Colab

EXECUTIVE SUMMARY

Investment risk assessment is one of the most important activities in financial decision-making. Investors and organizations rely on financial indicators to understand whether a company is financially stable or risky. Traditional methods of investment risk assessment are usually performed manually through financial ratio analysis and expert judgment. However, such methods are often time-consuming, subjective, and difficult to apply to large datasets.

The purpose of this project is to develop a machine learning-based investment risk prediction system capable of identifying the financial risk level of companies using financial indicators. The project uses the Polish Companies Bankruptcy Dataset, which contains financial variables such as profitability ratios, leverage indicators, liquidity measures, and operational metrics.

Three machine learning models were implemented in this project:

1. Logistic Regression

2. Decision Tree
3. Random Forest

Among these models, Random Forest achieved the highest accuracy of approximately 96.63%, making it the best-performing model for this study.

The project also integrates Explainable Artificial Intelligence (XAI) using SHAP to improve interpretability. SHAP analysis helps identify the contribution of each financial feature towards the final prediction.

A risk scoring system ranging from 0 to 100 was also developed, enabling companies to be categorized into Low Risk, Medium Risk, and High Risk groups.

The findings of this project demonstrate that machine learning techniques can effectively support financial decision-making by providing accurate and interpretable investment risk predictions.

CHAPTER 1 – INTRODUCTION

1.1 Introduction to Investment Risk Assessment

Investment risk assessment is one of the most important activities in the field of finance and investment management. Investors, banks, organizations, and financial analysts constantly evaluate companies before making investment decisions. The main purpose of risk assessment is to determine whether a company is financially stable or likely to face financial difficulties in the future.

Traditionally, investment decisions were taken manually by analyzing financial reports, balance sheets, income statements, and various profitability ratios. Financial experts studied the company's performance over multiple years and then estimated the level of investment risk associated with that company. Although these methods are useful, they are often time-consuming, subjective, and difficult to implement on a large scale.

In recent years, the rapid growth of financial data and technological advancements has created the need for automated systems capable of analyzing huge amounts of financial information. Machine learning has emerged as one of the most effective technologies for solving such problems because it can identify hidden patterns in financial datasets and generate accurate predictions.

Machine learning models can process thousands of financial records within seconds and provide faster and more reliable outputs compared to manual methods. These models are capable of learning relationships between financial indicators and company risk levels.

The present project focuses on developing a machine learning-based investment risk assessment system that predicts the financial risk level of companies using financial indicators obtained from a secondary dataset.

1.2 Background of the Study

The financial industry has undergone significant transformation due to the integration of data analytics and artificial intelligence technologies. Financial organizations today generate large amounts of structured and unstructured data. Managing and analyzing this data manually has become increasingly difficult.

Investment risk prediction is an important task because incorrect investment decisions can lead to major financial losses. Investors need systems that can:

- Predict risky companies accurately
- Analyze financial indicators quickly
- Reduce dependency on manual analysis
- Improve transparency in decision-making

Traditional methods mainly rely on statistical analysis and expert opinions. However, such methods may not always identify complex nonlinear relationships among financial variables.

Machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest can automatically learn from historical financial data and generate predictions based on learned patterns.

The integration of Explainable Artificial Intelligence (XAI) further improves the transparency of machine learning models by helping users understand how the prediction was generated.

1.3 Need for the Project

The need for automated investment risk assessment systems has increased significantly in recent years due to several reasons.

Increasing Financial Data

Financial institutions generate large amounts of data daily. Manual analysis of such large datasets becomes extremely difficult and inefficient.

Faster Decision-Making

Investors require quick financial insights for making timely investment decisions. Automated machine learning systems can produce predictions much faster than traditional methods.

Reduction of Human Error

Manual financial analysis may introduce subjectivity and inconsistencies. Machine learning models reduce human dependency and improve consistency.

Demand for Explainability

Modern financial systems require transparency. Explainable AI techniques such as SHAP help users understand why a particular prediction was generated.

Growth of Artificial Intelligence in Finance

The finance industry is increasingly adopting AI-driven systems for risk analysis, fraud detection, customer analytics, and investment prediction.

1.4 Problem Statement

Traditional investment risk assessment techniques are time-consuming, subjective, and difficult to scale for large financial datasets. Most manual systems depend heavily on human expertise and may fail to identify hidden relationships among financial indicators.

There is a need for a machine learning-based framework that can:

- Predict investment risk accurately
- Process large-scale financial data
- Provide interpretable results
- Reduce manual effort
- Support real-time financial decision-making

The project aims to solve these challenges by developing an automated investment risk prediction system using machine learning and explainable AI.

1.5 Objectives of the Project

The primary objectives of the project are as follows:

- 1: To Develop a Machine Learning-Based Investment Risk Prediction System
- 2: To Compare Multiple Machine Learning Algorithms
- 3: To Implement Explainable AI

1.6 Scope of the Project

The scope of this project includes financial risk prediction using machine learning techniques.

The project covers:

- Data preprocessing
- Financial indicator analysis
- Machine learning classification
- Model evaluation

- Explainable AI implementation
- Risk score generation

The developed system can be extended for:

- Banking systems
- Portfolio management
- Credit scoring systems
- Real-time investment advisory platforms
- Financial analytics dashboards

1.7 Limitations of the Project

Although the project achieves strong performance, certain limitations exist.

Dataset Imbalance

The dataset contains more low-risk companies than high-risk companies, which affects minority class prediction.

Limited Real-Time Integration

The current system works on historical financial datasets and does not directly integrate with live stock market APIs.

1.8 Advantages of the Proposed System

The proposed system offers several advantages compared to traditional manual analysis.

The proposed investment risk prediction system offers several advantages compared to traditional manual financial analysis methods. One of the major advantages is improved prediction accuracy, as machine learning models are capable of identifying hidden relationships and complex patterns within financial indicators that may not be easily recognized through manual analysis. This enables the system to generate more accurate and reliable predictions regarding company risk levels.

Another important advantage of the system is faster processing capability. The machine learning framework can analyze and process large financial datasets within a very short period of time, significantly reducing the time required for financial evaluation and decision-making.

The proposed system also reduces dependency on manual financial analysis. Traditional methods rely heavily on human expertise and interpretation, which may introduce inconsistencies and human errors. By automating the prediction process, the system minimizes manual effort and improves consistency in financial risk assessment.

In addition, the integration of SHAP-based explainable artificial intelligence improves interpretability and transparency. SHAP analysis helps users understand the contribution of individual financial indicators towards the final prediction, making the system more trustworthy and easier to interpret.

The system is also highly scalable and can be extended in the future to handle larger datasets, additional financial indicators, and real-time financial data. This makes the proposed framework suitable for practical deployment in modern financial environments.

1.9 Overview of Machine Learning Models Used

Logistic Regression

Logistic Regression is a statistical classification algorithm commonly used for binary classification tasks. It predicts probability values and classifies outputs into predefined categories.

Decision Tree

Decision Tree is a supervised learning algorithm that creates decision rules based on feature values.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve accuracy and reduce overfitting.

Among all implemented models, Random Forest achieved the highest accuracy.

1.10 Organization of the Report

The report is organized into six chapters.

- Chapter 1 explains the introduction and objectives.

- Chapter 2 discusses the finance industry and dataset profile.
- Chapter 3 presents the literature review.
- Chapter 4 explains the research methodology.
- Chapter 5 discusses implementation results and findings.
- Chapter 6 presents the conclusion and future scope.

CHAPTER 2 – INDUSTRY & COMPANY PROFILE

2.1 Introduction to the Finance Industry

The finance industry is one of the most important sectors in the global economy. It plays a major role in economic development by managing money, investments, loans, insurance, and financial transactions. The industry supports businesses, governments, and individuals by providing financial services that help in economic growth and wealth generation.

The finance industry includes several sectors such as:

- Banking
- Investment management
- Insurance
- Stock markets
- Financial technology services
- Credit and lending institutions

Modern financial systems are highly data-driven. Every transaction, investment activity, and financial operation generates large amounts of data. Managing and analyzing this data manually is extremely difficult. Therefore, organizations are increasingly adopting machine learning and artificial intelligence technologies for financial analysis.

2.2 Importance of Investment Risk Assessment

Investment risk assessment is an essential activity in finance because investors need to understand whether investing in a company is financially safe or risky.

A wrong investment decision can lead to serious financial consequences for both individuals and organizations. Poor investment choices may result in financial losses, bankruptcy, weak portfolio performance, and reduced investor confidence. Because of these risks, financial institutions and investment firms perform detailed risk analysis before approving loans, investments, or credit facilities. Investment risk assessment plays a significant role in identifying financially stable companies and avoiding organizations that may face financial distress in the future. Effective risk assessment also helps investors improve portfolio management, reduce uncertainty in decision-making, and make informed financial decisions based on data and financial analysis.

2.3 Evolution of Financial Technology (FinTech)

Financial Technology, commonly known as FinTech, refers to the integration of modern digital technologies into financial services and operations. FinTech has significantly transformed the traditional finance industry by improving the efficiency, accessibility, speed, and accuracy of financial processes. Earlier, most financial activities such as banking, payments, investments, and financial analysis were performed manually or through traditional systems, which were often time-consuming and less efficient. However, with the advancement of technology, financial institutions have increasingly adopted digital solutions to automate and improve financial services.

FinTech has introduced several innovative services and platforms that have changed the way individuals and organizations manage financial activities. These include online banking systems that allow customers to access banking services remotely, digital payment platforms that support instant transactions, AI-based financial analytics systems, automated trading platforms used in stock markets, cryptocurrency and blockchain-based financial services, and robo-advisors that provide automated investment recommendations.

2.4 Role of Machine Learning in Finance

Machine learning is widely used in the finance industry because of its ability to analyze large amounts of data efficiently.

Machine learning systems can learn from historical financial data and identify relationships between financial indicators and business outcomes.

Applications of machine learning in finance include:

Fraud Detection

Machine learning models identify suspicious financial transactions and prevent fraud.

Credit Scoring

Banks use machine learning to determine whether a customer is eligible for loans.

Stock Market Prediction

Machine learning algorithms analyze stock market trends and forecast future price movements.

Bankruptcy Prediction

Companies facing financial distress can be identified using predictive models.

Investment Risk Analysis

Machine learning systems classify companies based on financial risk levels.

The present project focuses specifically on investment risk analysis using financial indicators.

2.5 Company Profile – Polish Companies Bankruptcy Dataset

The dataset used in this project is the Polish Companies Bankruptcy Dataset, which is one of the most widely used datasets in research related to bankruptcy prediction, financial risk analysis, investment forecasting, and machine learning-based financial classification systems. The dataset has been extensively utilized by researchers and data analysts for studying company financial health and developing predictive models capable of identifying financially risky organizations.

The dataset contains financial information of companies collected over multiple years and includes a large number of financial indicators that describe the operational and financial condition of different firms. These financial indicators provide valuable insights into the profitability, liquidity, leverage, and overall stability of companies. The dataset also contains bankruptcy classification labels that identify whether a company is financially stable or likely to face bankruptcy-related issues.

Several important financial indicators are included in the dataset, such as profitability ratios, liquidity measures, leverage ratios, operational efficiency indicators, and debt-related metrics. These indicators are essential for evaluating company performance and determining investment risk levels. Profitability indicators help measure the company's earning capability, liquidity ratios evaluate short-term financial stability, and leverage indicators analyze the company's dependency on debt and borrowed funds.

The dataset is highly suitable for machine learning applications because it contains structured numerical financial data along with predefined class labels required for supervised learning tasks. The availability of both financial features and classification labels makes it ideal for training and evaluating machine learning models such as Logistic Regression, Decision Tree, and Random Forest. The dataset also provides sufficient complexity and feature diversity, allowing machine learning algorithms to identify hidden patterns and relationships among financial indicators and company risk levels.

2.6 Financial Indicators Used in the Project

Financial indicators are numerical values that represent the financial condition, operational efficiency, and overall stability of a company. These indicators are widely used by investors, financial analysts, and organizations to evaluate company performance and determine whether a company is financially strong or risky for investment purposes. Financial indicators provide measurable insights into different aspects of business performance such as profitability, liquidity, leverage, operational efficiency, and earnings stability.

The present project utilizes several important financial indicators from the dataset to predict investment risk using machine learning algorithms. Each indicator contributes differently towards understanding the financial health of a company and plays an important role in the prediction process.

One of the major indicators used in the project is Net Income to Stockholder's Equity. This indicator measures the profitability generated relative to shareholder investment and helps determine how efficiently a company is utilizing shareholder funds to generate earnings. Companies with higher profitability are generally considered financially stable and less risky for investment.

Another important indicator is Borrowing Dependency, which measures the extent to which a company depends on borrowed funds and external liabilities for its operations. A higher borrowing dependency indicates greater financial obligations and increased investment risk because the company may face difficulties in repaying debts during financial instability.

The project also considers the Degree of Financial Leverage (DFL), which measures the sensitivity of company earnings to changes in operating income. Companies with high financial leverage are more exposed to financial risk because even small changes in operating income can significantly affect overall profitability.

2.7 Industry Challenges in Financial Risk Prediction

Financial risk prediction systems face several challenges due to the complexity and dynamic nature of financial environments. Modern financial institutions generate and process massive amounts of data every day, making financial analysis increasingly difficult through traditional manual methods. Developing accurate and reliable financial prediction systems therefore requires handling multiple technical and analytical challenges effectively.

One of the major challenges is the management of large-scale financial data. Financial organizations continuously generate huge volumes of transactional records, financial statements, investment data, customer information, and market-related data. Processing and

analyzing such large datasets efficiently requires advanced computational techniques and machine learning models capable of handling high-dimensional data.

Another significant challenge is class imbalance in financial datasets. In most financial risk prediction datasets, the number of financially stable companies is much higher than the number of bankrupt or high-risk companies. This imbalance affects machine learning performance because models tend to become biased towards the majority class, making it difficult to accurately identify minority high-risk cases. As a result, models may achieve high overall accuracy while still failing to predict risky companies effectively.

Market volatility also creates major challenges for financial prediction systems. Financial markets are highly dynamic and are influenced by several factors such as economic conditions, government policies, inflation, political events, global crises, and investor behavior. Rapid changes in market conditions can affect financial indicators and make prediction models less stable over time.

2.8 Explainable AI in Finance

Traditional machine learning models are often considered black-box systems because their internal decision-making process is difficult to interpret and understand. Although these models may provide highly accurate predictions, users are often unable to determine why a particular prediction was generated or which financial factors influenced the final output. In financial applications, transparency and interpretability are extremely important because investors and organizations need clear explanations before making important financial decisions.

Explainable Artificial Intelligence (XAI) improves transparency by helping users understand the reasoning behind machine learning predictions. It provides insights into why a prediction was generated, which financial features contributed the most, and how different financial indicators influence the final risk classification. This improves user trust and makes the prediction system more reliable and understandable.

2.9 Future Trends in Financial Analytics

The future of financial analytics is highly dependent on advancements in artificial intelligence, machine learning, and data science technologies. As financial markets continue to generate massive amounts of data, organizations are increasingly adopting intelligent systems to improve decision-making, reduce risk, and automate financial operations. Traditional financial

analysis methods are gradually being replaced by data-driven systems capable of generating faster and more accurate predictions.

Several important trends are shaping the future of financial analytics. Real-time investment prediction systems are becoming increasingly popular because they allow investors to analyze market conditions and generate predictions instantly. AI-powered financial advisors and robo-advisory platforms are also transforming the investment industry by providing automated financial guidance and portfolio recommendations.

Automated portfolio management systems are helping organizations optimize investments using machine learning algorithms and predictive analytics. In addition, blockchain-based financial systems and cryptocurrency platforms are introducing new approaches for secure and transparent financial transactions.

2.10 Summary

This chapter discussed the finance industry, the role of machine learning in finance, the dataset used in the project, and major industry challenges. The chapter also highlighted the importance of explainable AI and financial indicators in investment risk prediction.

CHAPTER 3 – LITERATURE REVIEW

3.1 Introduction

A literature review is an important phase of research that helps in understanding previous studies and existing work related to the project topic. It provides a strong foundation for the research by analyzing how other researchers approached similar problems, which methodologies they used, and what results they achieved. A literature review also helps identify the strengths and limitations of existing systems and supports the development of improved research solutions.

The main purpose of conducting a literature review is to identify existing research studies related to financial risk prediction and machine learning applications in finance. It also helps in studying the methodologies and algorithms used by previous researchers, understanding research gaps and limitations, comparing findings from different studies, and developing improved and more efficient solutions for the proposed problem.

This chapter reviews several research papers and studies related to financial risk prediction, bankruptcy prediction, machine learning techniques in finance, and Explainable Artificial Intelligence (XAI). The review helps in understanding how machine learning models are used for financial analysis and how explainability techniques improve transparency and interpretability in prediction systems.

3.2 Traditional Financial Risk Prediction Approaches

Initial financial risk prediction systems relied heavily on traditional statistical methods and manual financial analysis for evaluating company performance and investment risk. Financial analysts and researchers mainly used techniques such as Logistic Regression, Linear Regression, and statistical probability models to study financial data and predict the likelihood of bankruptcy or financial distress. These methods were widely accepted because they were relatively simple to implement, mathematically interpretable, and easy to understand.

Although these traditional approaches provided useful insights into financial stability, they had several limitations. Most statistical models were only capable of handling linear relationships between financial variables and often struggled to identify complex nonlinear patterns present in large financial datasets. As financial systems became more complex and data volumes increased, the effectiveness of these traditional methods started decreasing.

3.3 Machine Learning-Based Financial Risk Prediction

Machine learning has significantly improved financial risk analysis by automating prediction and decision-making tasks. Unlike traditional statistical methods, machine learning algorithms are capable of learning complex relationships directly from historical financial data and generating predictions with minimal human intervention. This advancement has enabled financial institutions and researchers to process large-scale financial datasets more efficiently and accurately.

Researchers have implemented several machine learning algorithms for financial risk prediction, including Decision Trees, Random Forest, Support Vector Machines, Neural Networks, and Gradient Boosting Models. These algorithms have shown better performance compared to traditional statistical approaches because they can handle nonlinear relationships, large feature sets, and complex financial patterns more effectively.

Machine learning models demonstrated higher prediction accuracy in tasks such as bankruptcy prediction, credit scoring, fraud detection, and investment risk assessment. These systems are

capable of identifying hidden patterns and relationships within financial indicators that may not be easily recognized through manual analysis or conventional statistical techniques.

3.4 Random Forest in Financial Prediction

Random Forest became one of the most widely used machine learning algorithms for financial prediction and risk assessment tasks. It is an ensemble learning algorithm that combines multiple decision trees to improve prediction performance and reduce the limitations associated with individual decision tree models. Due to its robustness and high accuracy, Random Forest has gained significant popularity in financial analytics and bankruptcy prediction research.

Researchers found that Random Forest effectively reduces overfitting by averaging the predictions of multiple decision trees instead of relying on a single model. This improves the generalization capability of the system and allows the model to perform better on unseen financial data. The algorithm also improves classification accuracy by identifying complex patterns and relationships among financial indicators.

3.5 Explainable AI in Finance

Although machine learning models significantly improve prediction accuracy, many financial systems also require transparency and interpretability in their decision-making process. In financial applications, organizations and investors need to understand why a particular prediction was generated before making important financial decisions. Traditional machine learning models are often considered black-box systems because their internal operations are difficult to interpret.

To address this issue, Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME were introduced to improve the understanding of machine learning predictions. These techniques help users analyze how individual features influence the final output generated by the model. Explainable AI improves transparency and allows financial analysts to trust automated prediction systems more confidently.

3.6 Studies Related to Financial Indicators

Previous research studies identified several important financial indicators that significantly affect investment risk and financial stability. Researchers analyzed different types of financial ratios and performance metrics to understand how they influence company risk levels and bankruptcy prediction. These financial indicators are widely used in investment analysis because they provide insights into the operational efficiency and financial health of organizations.

Some of the most important indicators identified in previous studies include profitability ratios, liquidity measures, leverage indicators, earnings stability, and debt ratios. Profitability ratios help evaluate the earning capability of a company, while liquidity measures determine the ability of the company to meet short-term financial obligations. Leverage indicators and debt ratios analyze the company's dependence on borrowed funds and financial liabilities. Earnings stability reflects the consistency of company performance over time.

3.7 Research Gaps Identified

Although previous research studies achieved good results in financial risk prediction, several limitations still remain in existing systems. Many researchers primarily focused on improving prediction accuracy while ignoring important aspects such as interpretability, real-time capability, and practical usability. These limitations create challenges in applying machine learning systems effectively in real-world financial environments.

One of the major limitations identified in previous studies is the lack of interpretability. Many machine learning models function as black-box systems, where users cannot clearly understand why a particular prediction was generated. In financial applications, transparency is extremely important because investors and organizations require proper explanations before making critical financial decisions.

Another limitation is the limited real-time capability of many existing systems. Several research models were designed only for offline prediction and were unable to support real-time financial analysis using continuously changing market data. This reduces the practical applicability of such systems in dynamic financial environments.

3.8 Proposed Research Contribution

The present project attempts to address these limitations by:

- Implementing multiple machine learning models
- Integrating SHAP explainability
- Generating risk scores between 0 and 100
- Comparing model performance
- Analyzing feature importance

3.9 Summary

This chapter reviewed previous research related to machine learning-based financial risk prediction and explainable AI. The literature review helped identify research gaps and provided the foundation for developing the proposed system.

CHAPTER 4 – RESEARCH METHODOLOGY

4.1 Introduction

Research methodology explains the procedures, techniques, tools, and approaches used for developing the proposed investment risk prediction system. It provides a structured framework for conducting the research and describes the step-by-step process followed during the implementation of the project. A well-defined research methodology helps ensure that the project is systematic, reliable, and capable of achieving the desired objectives.

This chapter discusses the complete workflow followed for developing the machine learning-based investment risk assessment system. It includes dataset collection and selection, data preprocessing techniques, machine learning model development, performance evaluation methods, and the implementation of Explainable Artificial Intelligence techniques. The chapter also explains how financial data was prepared, how different machine learning algorithms were trained and tested, and how model predictions were interpreted using SHAP analysis.

4.2 Research Design

The project follows a secondary research methodology because the dataset used is publicly available.

The overall workflow includes:

1. Dataset collection
2. Data preprocessing
3. Feature extraction
4. Model training
5. Prediction
6. Model evaluation
7. Explainability analysis
8. Risk score generation

4.3 Dataset Collection

The project uses the Polish Companies Bankruptcy Dataset for developing and evaluating the investment risk prediction system. The dataset contains financial records of companies along with class labels that indicate the financial condition and risk level of each company. It includes various financial indicators related to profitability, liquidity, leverage, operational efficiency, and overall financial performance.

The dataset was selected for this project because it contains large-scale financial data that is highly suitable for machine learning applications. The availability of multiple financial indicators allows machine learning algorithms to identify relationships between financial variables and company risk levels effectively. In addition, the dataset is widely used in research related to bankruptcy prediction, financial distress analysis, and investment risk assessment, making it a reliable and academically relevant source for the project.

The structured nature of the dataset and the presence of predefined classification labels make it appropriate for supervised machine learning tasks such as classification and risk prediction.

4.4 Data Preprocessing

Data preprocessing is one of the most important phases of machine learning.

Raw financial datasets often contain:

- Missing values
- Duplicate records
- Inconsistent data
- Noise

The following preprocessing steps were performed.

Several data preprocessing techniques were applied to prepare the dataset for machine learning implementation. Data preprocessing is an important step because raw financial datasets often contain missing values, inconsistent records, and unstructured information that can negatively affect model performance.

The first step involved handling missing values present in the dataset. Missing data entries were either removed or replaced appropriately to ensure data consistency and improve the reliability of the prediction models. Proper handling of missing values helped prevent errors during model training and evaluation.

The next step involved data cleaning, where irrelevant, duplicate, and inconsistent data entries were identified and removed from the dataset. This improved the overall quality of the financial data and ensured that only meaningful information was used for analysis.

Feature scaling was also performed to normalize numerical values present in the dataset. Since financial indicators may exist on different scales, normalization helps maintain consistency and improves machine learning performance by preventing certain features from dominating the prediction process.

4.5 Machine Learning Algorithms Used

Logistic Regression

Logistic Regression is a supervised learning algorithm used for classification problems.

Advantages:

- Simple implementation
- Good baseline model
- Interpretable outputs

Limitations:

- Limited ability to capture nonlinear relationships

Decision Tree

Decision Tree creates decision rules using feature values.

Advantages:

- Easy visualization
- Handles nonlinear relationships

Limitations:

- Overfitting issues

Random Forest

Random Forest combines multiple decision trees.

Advantages:

- High accuracy
- Reduced overfitting
- Better generalization
- Stable predictions

Random Forest achieved the best results in this project.

4.6 Model Training Process

The machine learning models used in this project were implemented and trained using the Scikit-learn library in Python. Scikit-learn provides efficient tools and algorithms for machine learning model development, training, and evaluation. It was used to implement Logistic Regression, Decision Tree, and Random Forest models for investment risk prediction.

The training process involved feeding the training dataset into the machine learning algorithms so that the models could learn patterns and relationships from historical financial data. During training, the models analyzed various financial indicators such as profitability ratios, leverage measures, and liquidity indicators to understand how these features influence investment risk levels.

After learning the relationships among financial variables, the models generated predictive classification systems capable of categorizing companies according to their financial risk. The outputs generated by the models were then evaluated using different performance metrics such as accuracy, precision, recall, and F1-score.

4.7 Performance Evaluation Metrics

Several performance evaluation metrics were used to measure the effectiveness and accuracy of the machine learning models implemented in this project. These metrics help analyze how well the models classify companies into different investment risk categories and provide a detailed understanding of prediction performance.

One of the primary evaluation metrics used was accuracy, which measures the percentage of correctly classified instances out of the total predictions made by the model. Accuracy provides an overall measure of model performance and helps compare different machine learning algorithms.

Precision was also used as an important evaluation metric. Precision measures the correctness of positive predictions made by the model and indicates how many predicted positive cases were actually correct. Higher precision values indicate fewer false positive predictions.

Another important metric used in the project is recall. Recall measures the ability of the model to correctly identify actual positive cases, especially high-risk companies. It indicates how effectively the system detects relevant cases from the dataset.

F1-score was also calculated to provide a balanced measure of model performance. The F1-score represents the balance between precision and recall and is particularly useful when dealing with imbalanced datasets, where one class contains significantly more instances than the other.

4.8 Explainable AI Using SHAP

SHAP analysis was implemented to improve interpretability.

SHAP helps explain:

- Which features influence predictions
- Whether features increase or decrease risk
- Relative importance of indicators

This improves trust and transparency.

4.9 Risk Score Generation

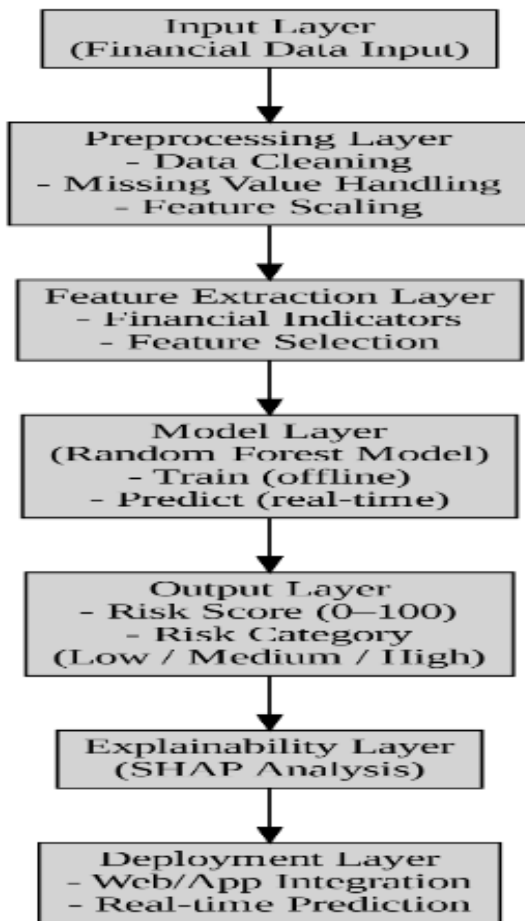
Instead of generating only class labels, the project converts prediction probabilities into numerical risk scores.

Risk score ranges:

- 0–33 : Low Risk
- 34–66 : Medium Risk
- 67–100 : High Risk

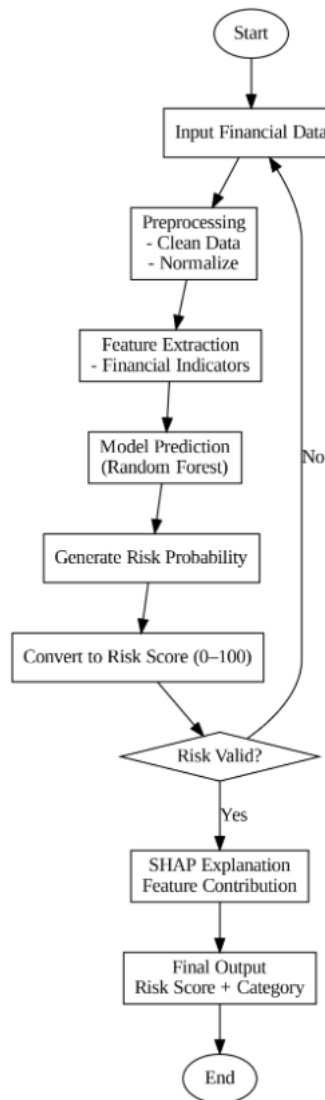
This makes the system more practical.

4.10 System Architecture



The system architecture illustrates the complete workflow from financial data input to prediction output.

4.11 Process Flow Diagram



The process flow diagram explains the sequence of operations performed by the proposed system.

4.12 Tools and Technologies Used

The project uses multiple tools and technologies.

Python

Used for model implementation and data analysis.

Pandas

Used for data manipulation.

NumPy

Used for numerical operations.

Scikit-learn

Used for machine learning implementation.

Matplotlib

Used for visualization.

Google Colab

Used as the development environment.

4.13 Summary

This chapter discussed the research methodology, dataset preprocessing, machine learning implementation, explainability analysis, and evaluation metrics used in the project.

CHAPTER 5 – DATA ANALYSIS & FINDINGS

5.1 Introduction

This chapter presents the implementation results, performance analysis, feature analysis, and major findings obtained from the proposed machine learning-based investment risk prediction system. The chapter focuses on evaluating the effectiveness of different machine learning models and analyzing how financial indicators influence investment risk prediction.

Three machine learning algorithms were implemented and evaluated in this project, namely Logistic Regression, Decision Tree, and Random Forest. Each model was trained using financial data from the Polish Companies Bankruptcy Dataset and tested to analyze its prediction capability and overall performance.

The primary purpose of this analysis is to identify the best-performing machine learning model for investment risk prediction and compare the strengths and limitations of each algorithm. In addition, the chapter studies the influence of different financial indicators such as profitability, leverage, liquidity, and earnings stability on the prediction process. The findings obtained from the analysis help understand how machine learning techniques can support accurate and interpretable financial risk assessment.

5.2 Model Accuracy Comparison

The machine learning models were evaluated using accuracy metrics.

Table 1 – Model Accuracy Comparison

Model	Accuracy
Logistic Regression	96.40%
Decision Tree	94.79%
Random Forest	96.63%

The Random Forest model achieved the highest accuracy among all models.



	Model	Accuracy
0	Logistic Regression	0.964076
1	Decision Tree	0.953079
2	Random Forest	0.967009

Figure 3 shows the comparison of model accuracy.

The results indicate that Random Forest provides better prediction capability due to its ensemble learning approach.

5.3 Logistic Regression Analysis

Logistic Regression served as the baseline classification model.

The model achieved:

- High precision for low-risk companies
- Strong overall accuracy
- Limited minority class prediction

The model struggled to identify high-risk companies because of dataset imbalance.

Advantages observed:

- Simpler implementation
- Faster training
- Good interpretability

Limitations observed:

- Reduced minority class recall
- Limited nonlinear learning capability

5.4 Decision Tree Analysis

Decision Tree classification produced moderate results.

Advantages:

- Easy visualization
- Interpretable decision rules
- Handles nonlinear data

Limitations:

- Lower generalization
- Overfitting tendency
- Reduced accuracy compared to Random Forest

The model achieved lower overall performance because single decision trees are highly sensitive to data variations.

5.5 Random Forest Analysis

Random Forest achieved the best overall performance among all the machine learning models implemented in this project. The algorithm is an ensemble learning technique that combines multiple decision trees to generate more accurate and stable predictions. By using multiple trees instead of relying on a single model, Random Forest improves prediction reliability and significantly reduces the problem of overfitting commonly associated with individual decision trees.

Several important observations were identified during the analysis of the Random Forest model. The model achieved the highest accuracy compared to Logistic Regression and Decision Tree, demonstrating its superior prediction capability for investment risk assessment. It also provided better prediction stability and improved robustness while handling financial datasets containing multiple financial indicators.

Another major advantage observed was the model's ability to effectively handle complex financial features and identify hidden relationships among profitability ratios, leverage indicators, liquidity measures, and earnings-related variables.

5.6 Classification Report Analysis

```
Logistic Regression
Accuracy: 0.9640762463343109
      precision    recall  f1-score   support

     0       0.97       0.99       0.98      1313
     2       0.56       0.20       0.29        51

 accuracy          0.96      1364
 macro avg          0.76       0.59       0.64      1364
 weighted avg       0.95       0.96       0.96      1364

-----
Decision Tree
Accuracy: 0.9479472140762464
      precision    recall  f1-score   support

     0       0.97       0.97       0.97      1313
     2       0.31       0.33       0.32        51

 accuracy          0.95      1364
 macro avg          0.64       0.65       0.65      1364
 weighted avg       0.95       0.95       0.95      1364

-----
Random Forest
Accuracy: 0.966275659824047
      precision    recall  f1-score   support

     0       0.97       1.00       0.98      1313
     2       0.78       0.14       0.23        51

 accuracy          0.97      1364
 macro avg          0.87       0.57       0.61      1364
 weighted avg       0.96       0.97       0.95      1364

-----
```

The classification report provides:

- Precision
- Recall
- F1-score
- Support values

The majority class achieved high precision and recall.

However, the minority class achieved lower recall values due to class imbalance.

This demonstrates one of the major challenges in financial risk prediction.

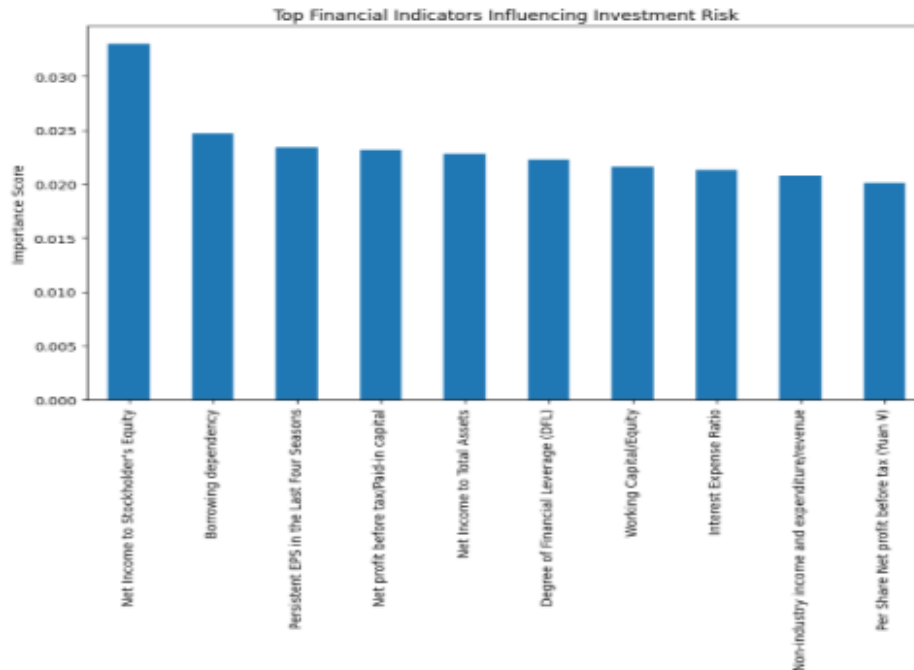
5.7 Feature Importance Analysis

Feature importance analysis was performed to identify which financial indicators contribute the most towards investment risk prediction. This analysis helps in understanding how different financial variables influence the behavior of the machine learning model and which features play a major role in determining whether a company is classified as low risk or high risk.

The analysis revealed that several financial indicators had a strong impact on prediction outputs. Some of the most influential features identified in the project include Net Income to Stockholder's Equity, Borrowing Dependency, Degree of Financial Leverage, Interest Expense Ratio, and Persistent Earnings Per Share (EPS). These indicators significantly affected the prediction capability of the machine learning models.

Net Income to Stockholder's Equity was found to be one of the strongest profitability-related indicators, reflecting how efficiently a company utilizes shareholder investment to generate earnings. Borrowing Dependency and Degree of Financial Leverage were important leverage-related indicators that measured the company's dependence on debt and financial liabilities. The Interest Expense Ratio represented the financial burden associated with debt servicing, while Persistent EPS indicated earnings consistency and overall financial stability.

5.8 Comparative Feature Analysis



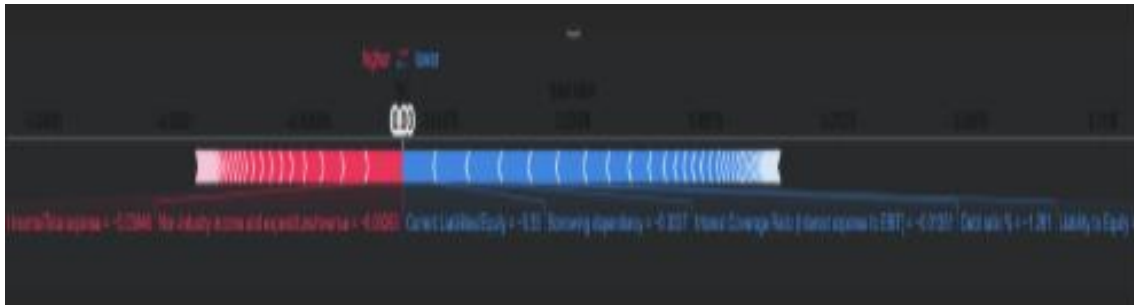
The comparative analysis confirms that profitability-related indicators consistently dominate the prediction process.

The model assigns high importance to:

- Equity-based profitability
- Debt dependency
- Financial leverage
- Earnings stability

This demonstrates that financially stable companies are classified as low risk.

5.9 Risk Score Distribution Analysis

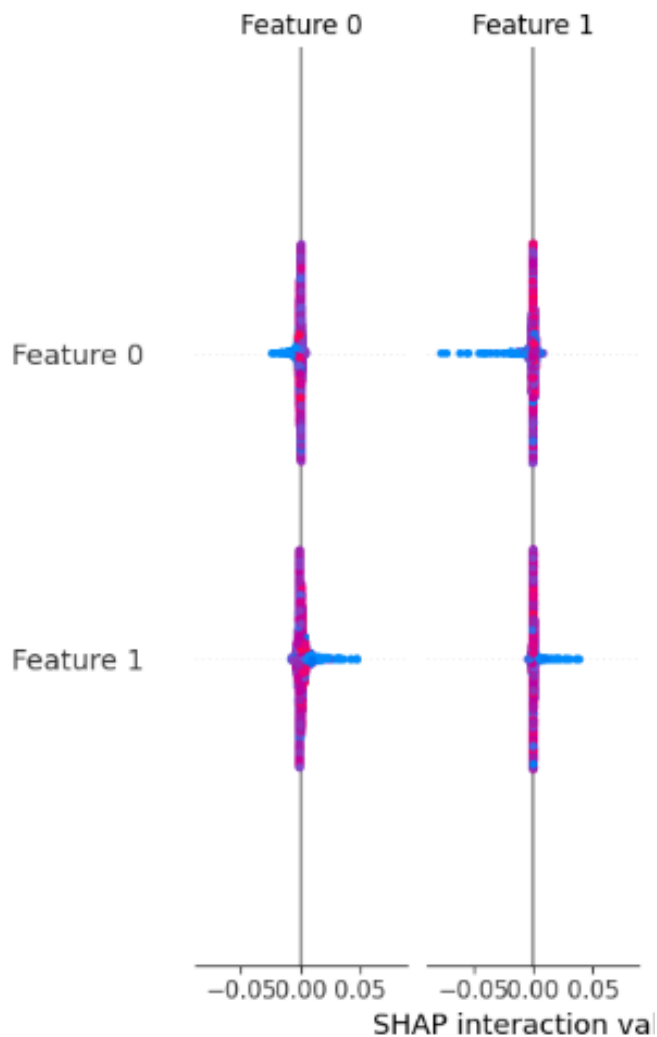


The risk score distribution analysis shows that the majority of companies in the dataset fall into the low-risk category, while only a smaller number of companies belong to the high-risk category. This indicates that the dataset is imbalanced, as financially stable companies are significantly more common than high-risk or bankrupt companies.

The presence of class imbalance affects machine learning performance because models tend to focus more on the majority of class during training. As a result, predicting minority high-risk companies becomes more challenging.

The developed risk scoring system improves the practical usability of the project by converting prediction probabilities into numerical risk scores instead of providing only categorical labels such as low-risk or high-risk. This allows users to better understand the level of financial risk associated with a company and supports more detailed investment analysis and decision-making.

5.10 SHAP Explainability Analysis



SHAP analysis explains the contribution of each financial feature.

The SHAP force plot indicates:

- Which features increase risk
- Which features reduce risk
- Relative contribution of indicators

This improves transparency and interpretability.

5.11 Major Findings

1. The analysis and implementation of the proposed investment risk prediction system resulted in several important findings. Among all the machine learning models implemented in the project, Random Forest achieved the highest accuracy and overall performance. Its ensemble learning approach enabled the model to generate stable and reliable predictions while effectively handling complex financial relationships present in the dataset.
2. The study also confirmed that financial indicators play a major role in determining investment risk levels. Variables related to profitability, leverage, liquidity, and earnings stability significantly influenced the prediction outputs generated by the machine learning models. Profitability and leverage indicators were identified as some of the strongest predictors of financial distress and investment risk.
3. Another important finding was the role of Explainable Artificial Intelligence in improving transparency and interpretability. The implementation of SHAP analysis helped explain the contribution of individual financial indicators towards prediction results, making the system easier to understand and more trustworthy for financial decision-making.

5.12 Summary

This chapter analyzed the performance of machine learning models and discussed major findings related to financial risk prediction.

CHAPTER 6 – CONCLUSION

The project successfully developed a machine learning-based investment risk prediction system using financial indicators and explainable AI.

Three machine learning models were implemented and compared:

- Logistic Regression
- Decision Tree
- Random Forest

Among these models, Random Forest achieved the highest accuracy of approximately 96.63%, making it the most suitable model for financial risk prediction.

The project also demonstrated the importance of explainable AI through SHAP analysis, which improved interpretability and trust in model predictions.

The risk scoring system developed in this project provides a practical way to classify companies into different risk categories.

Overall, the project demonstrates that machine learning can significantly improve investment risk assessment by providing accurate, scalable, and interpretable predictions.

Future Scope

Future improvements may include:

- Real-time financial data integration
- Deep learning models
- Improved handling of class imbalance
- Web-based deployment
- Portfolio recommendation systems

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- [5] K. Samara, M. Bennis, and S. K. Sharma, “Machine Learning for Financial Risk Prediction,” *Future Internet*, 2025.

APPENDIX

Appendix A – Tools Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- SHAP
- Google Colab

Appendix B – Dataset Information

Dataset Name: Polish Companies Bankruptcy Dataset

Appendix C – Screenshots

The screenshot shows a Google Colab notebook interface. The title bar reads "Risk assessment model". The left sidebar contains a "Table of contents" panel with a "+ Section" button. The main area displays three code cells. The first cell attempts to upload a file but shows a message: "Choose files No file chosen. Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable. Saving data.csv to data.csv". The second cell imports libraries: pandas, numpy, matplotlib.pyplot, and seaborn. The third cell imports sklearn modules: train_test_split, StandardScaler, RandomForestClassifier, DecisionTreeClassifier, LogisticRegression, classification_report, confusion_matrix, and accuracy_score. Below the code, the output shows the first five rows of the dataset as a table.

Bankrupt?	ROA(C) before interest and depreciation before interest	ROA(A) before interest and % after tax	ROA(B) before interest and depreciation after tax	Operating Gross Margin	Realized Sales Gross Margin	Operating Profit Rate	Pre-tax net Interest Rate	After- tax net Interest Rate	Non-industry expenditure,
0	1	0.370594	0.424389	0.405750	0.601457	0.601457	0.998969	0.796887	0.808809